

1. What is a computer network?
2. Discuss various types of networks topologies in computer network.
3. Also, discuss advantages and disadvantages of each topology.
4. What are the applications of Computer Networks?
5. What is OSI Model? Explain the functions, protocols and services of each layer?
6. Explain the following:-
 - I. LAN
 - II. MAN
 - III. WAN
7. What are the network components? What do you mean by client-server network?
8. Write some protocol names of each layer of TCP/IP model.
9. Explain protocol, layer, peer and interface using layered diagram.
10. Explain connection multiplexing, connectionless and connection-oriented services.
11. Explain OSI and TCP/IP model
12. Describe HDLC frame format.
13. How many types of HDLC frame there are? Explain control field formation of each of those.
14. When and why we use point –to-point connection?
15. Draw the ppp frame format. Explain each field.
16. Explain the diagram for bring a line up and down in a PPP communication.
17. ATM has how many cell formats? Explain the commonly used one.
18. Why Standardization is necessary? Explain with example
19. How does CSMA/CD work? Explain
20. What do mean by token? Explain IEEE 802.4 and 802.5.
21. Why not just one big LAN?
22. What devices are available for LAN's interconnection. Mention which layer they work.
23. Explain HUB and Bridge. Compare them.
24. Give a bridge learning example.
25. What will happen if there is a loop between two nodes?
26. What is spanning tree? How it helps bridge.

27. Compare and contrast between routers and bridge.
28. What do you mean by cut-through switching?
29. Compare and contrast between switch and bridge.
30. What is FDDI? How does it work?
31. What are the advantages of FDDI?
32. Draw the FDDI frame formation.
33. Differentiate between Ethernet and token ring.
34. If there is a network like 172.168.0.0/17, then determine the followings:
 - a. How many subnets are possible?
 - b. How many hosts are there?
 - c. What is the subnet mask of the network?
 - d. In the 1st subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?
 - e. In the 18th subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?
35. What are the physical media available for Ethernet, Fast Ethernet and Gigabit Ethernet?
Also note their maximum segment length.
36. **What leads to fast Ethernet?
37. What are the Ethernet and fast Ethernet and gigabit physical media types? Write down their maximum-segment-length.